

Data Flow Diagram

Date	05 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	Represents users who interact with the system. In this project, this is the Bank Loan Officer / Applicant who enters customer financial data and views the prediction results.
Rectangle with numbered header	Process	Represents the system operations that transform data. In this project, these are: 1. Data Preprocessing & Scaling (applying StandardScaler) 2. Model Inference Execution (predicting eligibility with the tuned XGBoost model) 3. Result Generation & Logging (saving predictions to the DB)
Rectangle (solid fill, no number)	Data Store	Represents where data is stored. In this project, this is the local SQLite database (smart_lender.db) which stores client records and prediction logs, and the source loan_prediction.csv file used for training.
Labeled arrow	Data Flow	Represents the movement of data between components. In this project, this includes: • Form Data Flow: Input fields passing from the UI to the Flask backend • Feature Flow: Transformed vectors passing to

		the model for prediction • Log Flow: Output predictions flowing to the database and final dashboard
--	--	--

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram



